Retrieval and De-duplication

Retrieval dispatches the configured query across 10 independent literature engines (arXiv, OpenAlex, Semantic Scholar, Crossref, PubMed, SovietRxiv, ChinaRxiv, Europe PMC, bioRxiv, and medRxiv). Each engine is an isolated adapter exposing a uniform `search(query) -> list[Record]` interface; engines that are keyless — arXiv, OpenAlex (Priem et al. 2022), Crossref (Hendricks et al. 2020), PubMed/Entrez (Sayers et al. 2022), SovietRxiv / RussiaRxiv, ChinaRxiv, Europe PMC, and bioRxiv/medRxiv — need no credentials, while Semantic Scholar (Kinney et al. 2023) uses a key when present. SovietRxiv is a translated archive of Soviet-era scientific preprints sourced from Math-Net.Ru and CyberLeninka (SovietRxiv Project 2026); ChinaRxiv serves translated Chinese preprints from ChinaXiv via the same unified API. Both retain original-language PDFs alongside each translation, and their polite rate-limit pool

(300/min vs 30/min anonymous) is activated by an optional `X-API-Email` header. Europe PMC is a keyless biomedical aggregator covering PubMed, PMC, patents, and preprints in a single search call. bioRxiv/medRxiv share one unified date-window + cursor API; unlike the other engines it is not a free-text search endpoint, so the adapter walks the date window page by page and keeps only records whose title and abstract match every query term client-side. Optional full-text resolution queries Unpaywall (Piwowar et al. 2018) for open-access locations. **Multiple dispatch degrades gracefully**: an engine that is disabled in the configuration, lacks a required key, or cannot reach the network returns a *skipped* status, and the run completes from the remaining engines plus the committed offline corpus.

Engine Details I

Each engine adapter follows a uniform contract: a module-level API URL constant, a pure `_parse_*` parser function, and a `search_*` entry point with pagination, retry, and graceful error handling. All functions accept an injectable `base_url` parameter for hermetic testing with `pytest-httpserver` — no engine hardcodes its URL inside the function body.

Engine Details I

Engine	Rate limit	Pagination	Auth
arXiv	3s between requests	100/page, offset	Keyless
Semantic Scholar	1 req/s (unauth.)	100/page, offset	Optional key
OpenAlex	Polite pool (mailto)	200/page, cursor	Keyless
Crossref	Polite pool (mailto)	1,000/page, offset	Keyless
PubMed	NCBI usage policy	retstart/retmax	Keyless
SovietRxiv	30/min (300/min polite)	1–100/page, cursor	`X-API-Email`
ChinaRxiv	30/min (300/min polite)	1–100/page, cursor	`X-API-Email`
Europe PMC	~10 req/s (undocumented hard limit)	Up to 1,000/page	Keyless
bioRxiv/medRxiv	No documented limit	100/page fixed, cursor	Keyless

Engine Details I

Every new search writes `output/data/retrieval_report.json`, a timestamp-free report that records each attempted, skipped, or failed source with fetched, new-record, and duplicate counts. A zero-result response is therefore distinguishable from a disabled adapter or an HTTP failure. The committed corpus predates that report contract, so this paper intentionally does not reconstruct source-specific counts from the merged corpus.

Canonical Identifier Hierarchy I

Heterogeneous records are reconciled by a **canonical identifier hierarchy** — DOI $>$ arXiv ID $>$ Semantic Scholar ID $>$ OpenAlex ID $>$ a stable digest of the normalized title. When two records share a canonical identifier they are merged, keeping the version with the most complete metadata (a count of non-None optional fields). The DOI is normalized: case-folded, resolver-prefix stripped, so the same paper returned by two engines under case/format-variant DOIs merges. For this run, 2260 records carry DOIs, 923 carry OpenAlex IDs, and 1 carry arXiv IDs. The de-duplicated corpus for this run holds $N = 2334$ records published across 2000–2026.

Relevance Filtering I

After de-duplication, a relevance filter drops papers whose title and abstract contain none of the configured relevance keywords (modafinil, armodafinil, provigil, wakefulness, narcolepsy, cognitive enhancement, alertness, sleep deprivation, vigilance, eugeroic). Keywords are matched case-insensitively; an empty keyword list is treated as no filter to avoid silently wiping the corpus. A year filter then excludes papers published before the configured start year (2000).